

## DATA SCIENCE PROJECT COODINATOR

### WEEK 1: PROJECT PLANNING AND STRATEGY DEVELOPMENT

#### Project Title:

# FreshPlan

## Meal Planner and Food Spoilage Prevention System

#### Prepared By:

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#### Role:

Individual Project Coordinator

#### Internship Duration:

07 September 2026 – 12 October 2026

#### Project Duration:

5 Weeks

#### Technology Focus:

Python | Pandas | NumPy | Matplotlib | Scikit-learn | Data Analysis

### EXECUTIVE SUMMARY

**FreshPlan** is a proposed data science project designed to reduce

household food waste by helping users understand how long stored food items are likely to remain usable and how they should be stored. Users can enter food items such as tomatoes, potatoes, carrots, rice, milk, bread, or other commonly stored foods. Based on the food item and storage condition, the system estimates its usable period and identifies items that need to be consumed first. The project also connects food-storage information with meal planning. When items are approaching their estimated expiry period, FreshPlan can suggest simple **breakfast, lunch, and dinner** options using those ingredients. Python-based data analysis techniques will be used to organize food dataset, analyze spoilage patterns, estimate spoilage risk, and generate useful insights. The project will be developed as an individual project over five weeks, covering planning, data collection, preprocessing, analysis, model development, testing, and final documentation.

## Project Goals

- Reduce avoidable food wastage.
- Help users understand food storage requirements.
- Estimate the usable period of stored food.
- Identify foods at higher spoilage risk.
- Prioritize food items that should be used first. Suggest breakfast, lunch, dinner ideas.
- Apply Python-based data science techniques. Present useful results through simple visualizations.

## Expected Deliverables

- Project plan
- Food dataset
- Data preprocessing
- Exploratory data analysis
- Spoilage-risk analysis
- Simple prediction/estimation component
- Meal recommendation component
- Data visualizations
- Final project report

## PROBLEM STATEMENT:

Food is often wasted because people forget:

- when food was stored
- how long it can normally be kept
- whether it should be refrigerated
- which items should be consumed first
- what meals can be prepared using food that may spoil soon

## Proposed Solution:

FreshPlan is a data-driven meal planning and food spoilage prevention system. The system allows users to enter stored food items and provides useful information such as:

- estimated usable duration
  - storage recommendation
  - approximate expiry/spoilage risk
  - priority items that should be consumed first
  - breakfast, lunch and dinner suggestions food combinations based on available items
- For example  
User enter : Tomato

| Information             | Example      |
|-------------------------|--------------|
| Food item               | Tomato       |
| Storage advice          | Refrigerator |
| Estimated usable period | 1-2 weeks    |
| Spoilage risk           | Medium       |
| Priority                | Use soon     |

## Data Science Role:

Mentioned here that Python-based data science techniques will be used for:

**Data collection → Data cleaning → Exploratory Data Analysis → Feature preparation → Rule/prediction logic → Visualization → Evaluation.**

## 4. Project Scope:

FreshPlan will focus on:

1. Food-item entry
2. Food database
3. Estimated shelf life
4. Storage recommendations
5. Spoilage-risk estimation
6. Priority-based food usage
7. Breakfast suggestions
8. Lunch suggestions
9. Dinner suggestions
10. Data analysis
11. Simple visualizations
12. Python-based processing

OUT OF SCOPE:

Real-time IoT refrigerator integration  
Hardware sensors  
Automatic camera-based food recognition  
Medical/nutritional diagnosis  
Exact laboratory-level spoilage prediction  
Large-scale commercial deployment

## 5. Project Objectives

**Primary Objective** To develop a data-driven food management system that helps users reduce food spoilage by providing estimated usable periods, storage guidance, spoilage-risk information, food-priority recommendations, and meal suggestions.

## Specific Objectives

- Build a structured food dataset.
- Analyze food storage and shelf-life patterns.
- Use Python for data preprocessing and analysis.
- Estimate the remaining usable period of food.
- Categorize foods according to spoilage risk.
- Prioritize foods that should be consumed earlier.
- Generate meal suggestions based on available foods.
- Create simple charts to communicate data insights.
- Evaluate the usefulness and accuracy of the system.

## Data Science Methodology

### 1. Data Collection

Create or collect a structured dataset containing:

- Food name
- Food category
- Storage method
- Typical shelf-life range
- Storage temperature/category
- Spoilage indicators
- Meal compatibility

Example:

| Food   | Category  | Storage           | Shelf Life    | Risk   |
|--------|-----------|-------------------|---------------|--------|
| Potato | Vegetable | Outside           | Dataset value | Low    |
| Milk   | Dairy     | Refrigerator      | Dataset value | High   |
| Bread  | Bakery    | Room/Refrigerator | Dataset value | Medium |

### 2. Data Preprocessing

Using **Pandas**:

- Remove duplicate records

Handle missing values  
Standardize food names  
Convert data types  
Organize categories  
Prepare the dataset for analysis

### 3. Exploratory Data Analysis

Using **Pandas + Matplotlib**, analyzes:

Which food categories spoil faster?  
Which foods have shorter usable periods?  
Which storage conditions are associated with longer usability?  
Which foods have higher spoilage risk?

### 4. Spoilage-Risk Estimation

You can create simple categories:

**Low Risk**  
**Medium Risk**  
**High Risk**

### 5. Priority System

**Use First → Use Soon → Normal → Long Storage**

### 6. Meal Recommendation

**Breakfast, Lunch, Dinner**

For example: Tomato + Bread + Onion

Possible suggestions: Breakfast → Tomato sandwich, Lunch → Tomato rice, Dinner → Tomato soup

### 7. Resources Required

## Hardware

Laptop/Desktop  
Internet connection  
Minimum required storage for project files

## Software

Python  
VS Code or Jupyter Notebook  
Required Python libraries

## Dataset

Food-item information  
Storage information  
Shelf-life estimates  
Spoilage-risk information  
Meal/ingredient mapping

**Project Coordinator / Developer:** Individual Student

## Responsibilities

Project planning  
Dataset preparation  
Python programming  
Data preprocessing  
Data analysis  
Model development  
Meal recommendation logic  
Visualization  
Testing  
Documentation  
Final presentation

## 8.PROJECT PHASE & GANTT CHART:

| PHASE | WEEK1 | WEEK2 | WEEK3 | WEEK4 | WEEK5 |
|-------|-------|-------|-------|-------|-------|
|-------|-------|-------|-------|-------|-------|

|                                   |  |  |  |  |  |
|-----------------------------------|--|--|--|--|--|
| PROJECT PLANNING                  |  |  |  |  |  |
| DATASET COLLECTION                |  |  |  |  |  |
| DATA CEANING                      |  |  |  |  |  |
| EXPLORATORY ANALYSIS              |  |  |  |  |  |
| SPOILAGE RISK COMPONENT           |  |  |  |  |  |
| MEAL RECOMMENDATION               |  |  |  |  |  |
| VISUALIZATION                     |  |  |  |  |  |
| TESTING                           |  |  |  |  |  |
| FINAL DOCUMENTATION/ PRESENTATION |  |  |  |  |  |

## 9. Risk Management Plan

Every data science project may face challenges related to data, time, technical development, and project scope. Since FreshPlan is being developed as an individual project, the main risks will be identified early and managed through proper planning and regular testing.

**Data Availability and Quality:** The project may face difficulty in obtaining sufficient and reliable food-related data, particularly information about shelf life, storage conditions, and spoilage patterns. To reduce this risk, the project will use a structured dataset and perform data cleaning before analysis. Missing, duplicate, or inconsistent values will be identified and handled during the preprocessing stage.

**Accuracy of Shelf-Life and Spoilage Estimates:** Food usability can vary depending on freshness, temperature, storage method, packaging, and other conditions. Therefore, the system will provide estimated usable periods and risk categories rather than claiming exact spoilage dates. The available data will be reviewed carefully, and the results will be presented as guidance rather than a guarantee of food safety.

**Limited Project Time:** As an individual project, completing all planned activities within five weeks may be challenging. This risk will be managed

by following the weekly project timeline and giving priority to the core features such as food data management, spoilage-risk estimation, food prioritization, and meal suggestions. Additional features will only be considered after the main functions are completed.

**Technical and Model-Related Issues:** Programming errors, data-processing problems, or poor model performance may occur during development. These issues will be reduced through step-by-step development, regular testing, and evaluation of the results. Simple and suitable Python-based methods will be preferred instead of unnecessarily complex models.

**Scope and Workload Management:** Since the project is developed by one person, adding too many features could make the project difficult to complete properly. The project scope will therefore remain focused on food spoilage prevention, storage guidance, priority-based food usage, and breakfast, lunch, and dinner recommendations. The development will follow the planned milestones to maintain quality and complete the project on time.

## 10. Success Criteria

The project will be considered successful if:

- Food items can be entered into the system.
- Relevant food information can be retrieved.
- Estimated usable periods can be displayed.
- Storage recommendations can be provided.
- Spoilage risk can be categorized.
- Foods can be prioritized based on urgency.
- Meal suggestions can be generated.
- Data analysis can produce meaningful insights.
- Visualizations clearly communicate results.
- The system performs reliably on test data.

## 11. Expected Outcomes

FreshPlan is expected to provide a simple data-driven approach to food management. The system will help users:

**Know → Store → Prioritize → Use → Reduce Waste.**

### Conclusion:

FreshPlan is a practical and socially useful data science project because it connects everyday household problems with Python-based data analysis and recommendation techniques. The first week focuses on building a strong foundation rather than immediately developing a complex application. By clearly defining the scope, assigning responsibilities, planning the data workflow, identifying risks, and preparing a timeline, the team can develop the project in a systematic and realistic manner.

The initial version will use simple and explainable recommendation rules. Later versions may include machine learning, image recognition, regional-language support, and waste-pattern prediction. This phased strategy makes the project achievable for a student team while still leaving room for meaningful innovation.